



INTRODUCTION

Data Structures & Algorithms

AIL 7002



Logistics

Instructors :

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Course web-page:

<https://ail7002iitd.github.io/>



Data-structures

Teaching assistants:

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TAs will help resolving any problems regarding the course :

coding, understanding a particular concept, assignments, etc.

Evaluations components

- Assignments : 30% (4)
- Quizzes (announced) : 10%
- Midterm Exam : 25%
- Major exam : 30%
- Attendance & Participation : 5%

Audit criteria: C, but a pass grade (30%) in major exam and pass grade (30%) in assignment is necessary for audit pass.

Programming Assignments

- 3 programming assignments
 - Must be done alone
 - late policy (penalty of 10% every day)
 - up to three days late

Logistics: Timings

- Timings

- Lets discuss!

- Piazza

Join class discussion group on Piazza (access code: ail7002)

https://piazza.com/iit_delhi/fall2026/ail7002

- Books

Data Structures and Algorithms in C++, 2nd Edition. Michael T. Goodrich,
Roberto Tamassia, David M. Mount

Algorithm Design, John Kleinberg and Eva Tardos.

Grading and Academic Integrity

■ Academic Integrity

- Cheating → negative penalty (and possibly more)

- Exception: if one person/team is identified as cheater without knowledge of the other team

- Non-cheater gets a zero

■ Collaboration is good!!! Cheating is bad!!! Who is a cheater?

Cheating vs Collaboration

So how do you draw the line between collaboration and cheating? Here's a reasonable set of ground rules.

The Kyunki Saas Bhi Kabhi Bahu Thi Rule: This rule says that you are free to meet with fellow students(s) and discuss assignments with them. Writing on a board or shared piece of paper is acceptable during the meeting; however, you should not take any written (electronic or otherwise) record away from the meeting. This applies when the assignment is supposed to be an individual effort or whenever two teams discuss common problems they are each encountering (inter-group collaboration). After the meeting, engage in a half hour of mind-numbing activity (like watching an episode of Kyunki Saas Bhi Kabhi Bahu Thi), before starting to work on the assignment. This will assure that you are able to reconstruct what you learned from the meeting, by yourself, using your own brain.

The Right to Information Rule: To assure that all collaboration is on the level, you must always write the name(s) of your collaborators on your assignment. This also applies when two groups collaborate.

- No coding by LLMs

LLMs for brainstorming, helping
with debugging



What is this course about ?

- Data-structures : how to organize data



What is this course about ?

- Data-structures : how to organize data
- Example case: Dictionary vs Thesaurus



What will I learn?

- How to argue formally about a data-structure.
- Complexity of each operation
- How to compare two data-structures



Discussion

Isn't this a 2nd year
undergraduate level course?



Topics

Data Structures

Arrays, Lists

Abstract Data Types, object-oriented concepts

Stacks, Queues

Trees: Binary trees, Balanced trees

Strings: Tries, Inverted Indexes

Sorting Algorithms

Hashing

Graphs

Algorithms

Greedy, Divide & Conquer, Dynamic Programming

Complexity Analysis & NP Completeness